

Bounded Almost-Sure Noise for Scalar Thresholding

Research Architecture Runtime
feature/lean-gated-operator-papers

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Abstract

For $\tilde{A}_T(x) = \mathbf{1}\{x + \xi \geq T\}$ with $|\xi| \leq \eta$ almost surely, pathwise application of the deterministic threshold-preservation theorem yields almost-sure output invariance outside the unstable band $[T - \eta, T + \eta)$. We explicitly **do not** claim the full Sparse Vector privacy theorem. Publication is gated on Lean `LEAN_FULL` for `Research.Operators.Threshold.BoundedNoise` (pathwise surrogate of a.s. bounded noise; `REAL.MATHLIB`).

1 Operator

Same equality convention $x \geq T$ passes. Noise is query-side only in the verified claim; threshold noise and $W = \xi - \zeta$ appear as portfolio candidates.

2 Main theorem

Theorem 1 (Bounded-noise threshold). *Let $\eta \geq 0$ and $|\xi| \leq \eta$ a.s. Then:*

1. *if $x \geq T + \eta$ then $\tilde{A}_T(x) = 1$ a.s.;*
2. *if $x < T - \eta$ then $\tilde{A}_T(x) = 0$ a.s.;*
3. *if $x \in [T - \eta, T + \eta)$ and $\eta > 0$, the random output need not be a.s. constant: under the two-point law $\xi \in \{-\eta, +\eta\}$ with positive mass on each atom, both outputs occur with positive probability on this band. (At the endpoint $x = T - \eta$, the $+\eta$ atom yields $x + \eta = T$, which passes; at interior points both $x \pm \eta$ land strictly on opposite sides of T . If $\eta = 0$ the unstable band is empty and part (3) is vacuous.)*

Proof. Parts (1)–(2): apply the deterministic theorem pathwise with $\varepsilon := \eta$. Part (3): for the two-point law on $\{\pm\eta\}$, the images $x \pm \eta$ realize both sides of the cut on the unstable band (same adversaries as deterministic sharpness, including the equality pass at $x = T - \eta$). $\square$

Theorem 2 (Noisy sharpness). *There exist admissible laws of ξ with $|\xi| \leq \eta$ a.s. such that if $x \in [T, T + \eta)$ then $\mathbb{P}(\tilde{A}_T = 0) > 0$, and if $x \in [T - \eta, T)$ then $\mathbb{P}(\tilde{A}_T = 1) > 0$. At $x = T + \eta$, $\tilde{A}_T = 1$ a.s. for every such ξ .*

3 Laplace utility identities (not a DP proof)

For $\xi \sim \text{Lap}(0, b)$ with margin $m = x - T$,

$$\mathbb{P}(x + \xi \geq T) = \begin{cases} 1 - \frac{1}{2}e^{-m/b}, & m \geq 0, \\ \frac{1}{2}e^{m/b}, & m < 0. \end{cases}$$

The repository checks this closed form at $m \in \{0, \pm \ln 2\}$ (with $b = 1$) and margin monotonicity.

This is **not** a differential-privacy accounting proof. Feedback limitation: `LAPLACE_CDF_IDENTITY_NOT_DP_PROOF`.

4 Limitations

Bounded noise $\neq$ Sparse Vector; no adaptive positive-release bound. Lean proof is a **pathwise** surrogate of a.s. bounded noise (`REAL_MATHLIB`); Laplace identities remain engineering checks, not a DP proof (`LAPLACE_CDF_IDENTITY_NOT_DP_PROOF`).

5 Verifier summary

Publication requires `LEAN_FULL` for `Research.Operators.Threshold.BoundedNoise` (`bounded_noise_preservation`, `bounded_noise_sharpness`). Certificate: `lean/certificates/thresholding/bounded-noise-threshold/`. Prior computational-only paper archived under `_archive/v0-computational/`.

References

References

- [1] Dwork and Roth, Algorithmic Foundations of Differential Privacy, 2014.
- [2] `implementation/src/operators/thresholding/`.